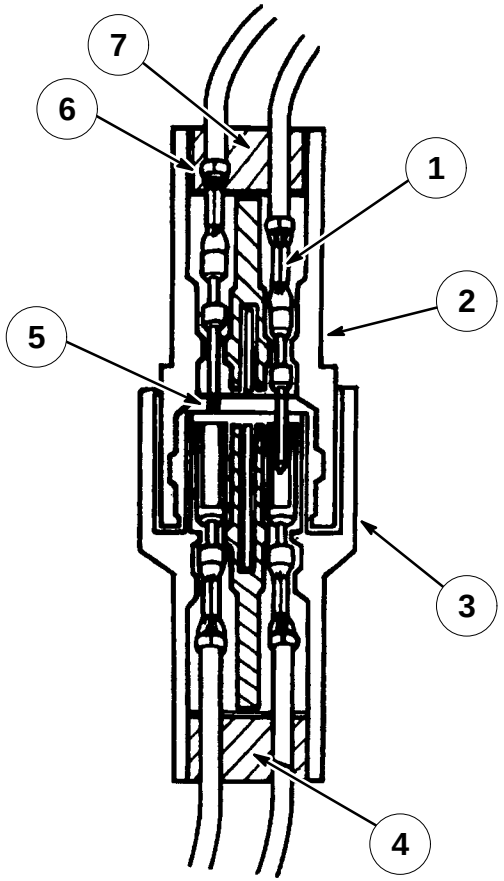
**Troubleshooting wiring harness and connector hidden concerns**

The following illustrations are known examples of wiring harness, splices and connectors that will create intermittent electrical concerns. The concerns are hidden and can only be discovered by a physical evaluation as shown in each illustration.

NOTE: Several components, such as the PCM, utilize gold plated terminals in their connections to the wiring harness. If those terminals need to be replaced, they must be replaced with a gold plated terminal.



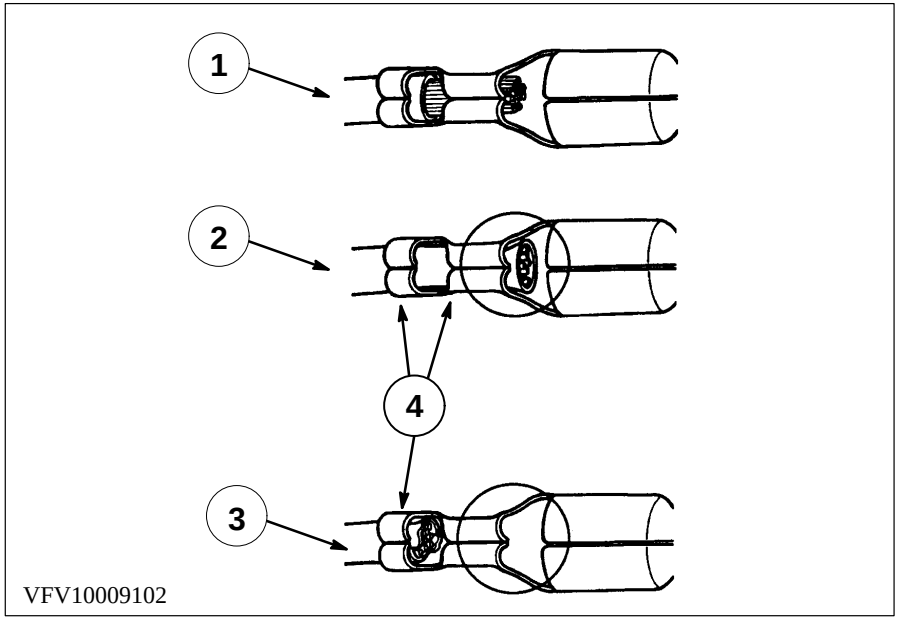
VFV10009101

**Terminal not properly seated**

1 = Locked terminal
2 = Male half
3 = Female half
4 = Seal
5 = Intermittent contact
6 = Unlocked terminal (Hidden by wire seal)
7 = Seal

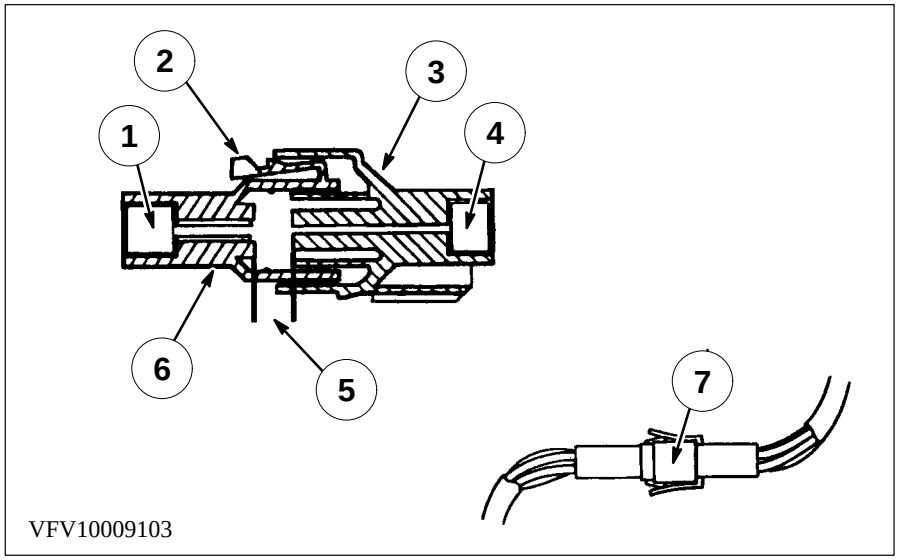
Check for unlocked terminals by pulling each wire at the end of the connector.

999




**Defective insulation stripping**

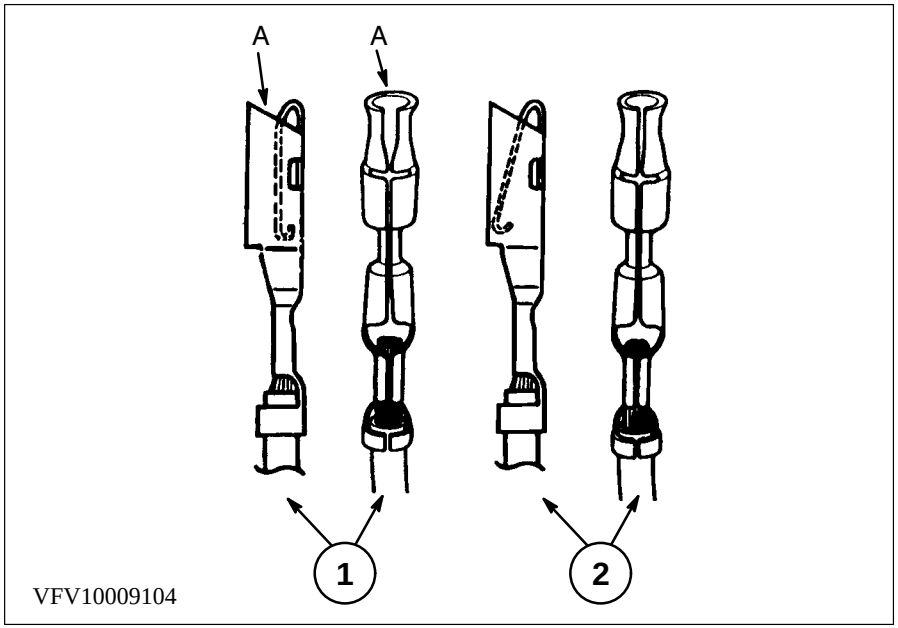
1 = Proper crimp
2 = Insulation not removed
3 = Wire strands missing
4 = Intermittent signals through pierced insulation




**Partially mated connectors**

1 = Seal
2 = Displaced tab
3 = Female half
4 = Seal
5 = Intermittent contact
6 = Male half
7 = Intermittent contact

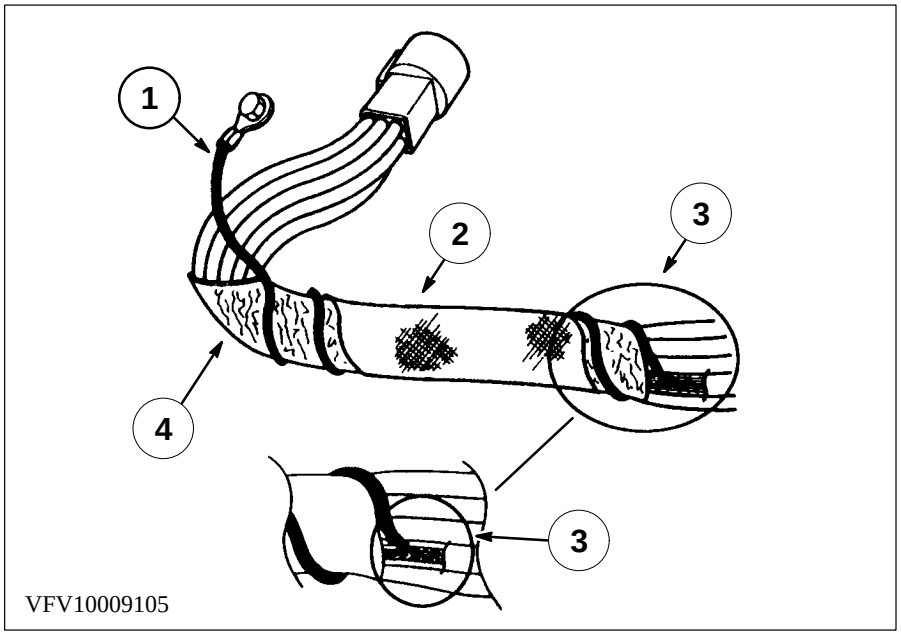
Lock may be displaced into an unlocked position; pull on the connector to verify the lock.




**Deformed (enlarged) female terminals**

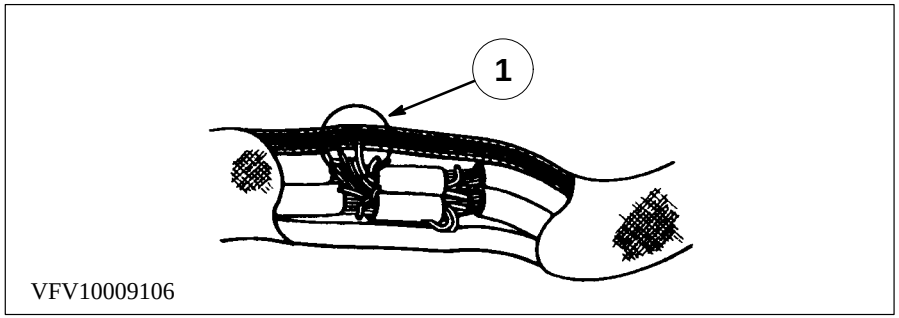
1 = Enlarged
2 = Normal

Any probe entering the terminal may enlarge the contact spring opening creating an intermittent signal. Insert the correct mating terminal (Location A) from the service kit and feel for a loose fit.




**Electrical short inside the harness**

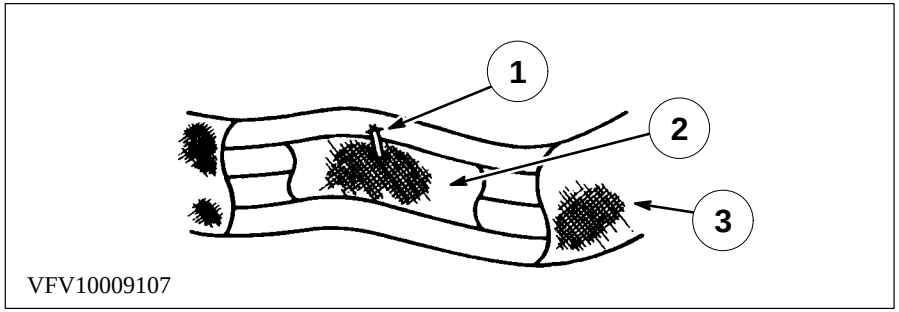
1 = Solder coated wire to ground
2 = Harness protective tape
3 = Intermittent short
Solder coated wire pierced through the insulation of another circuit
4 = Grounding foil




**Electrical short within the harness**

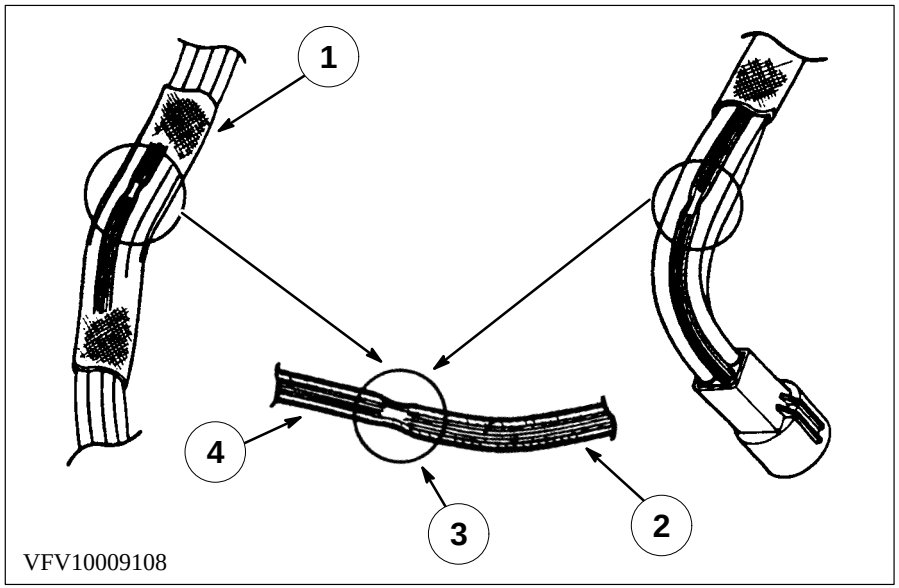
Splice tape removed

1 = Intermittent short




Splice covered

1 = Wire strand
2 = Splice tape
3 = Harness tape

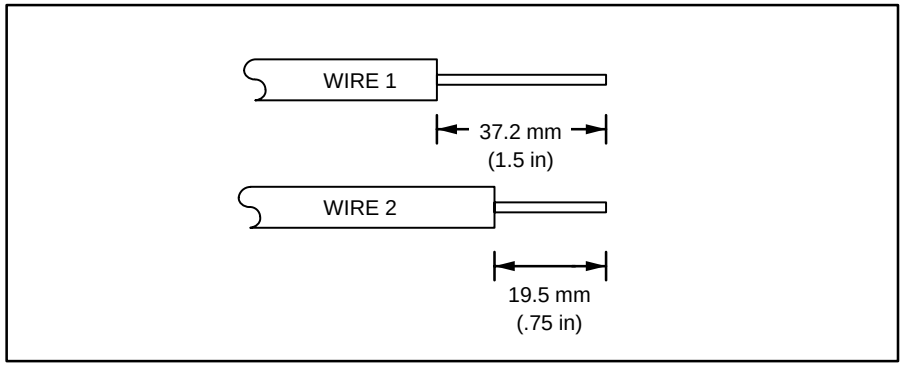



**Broken wire strands in harness**

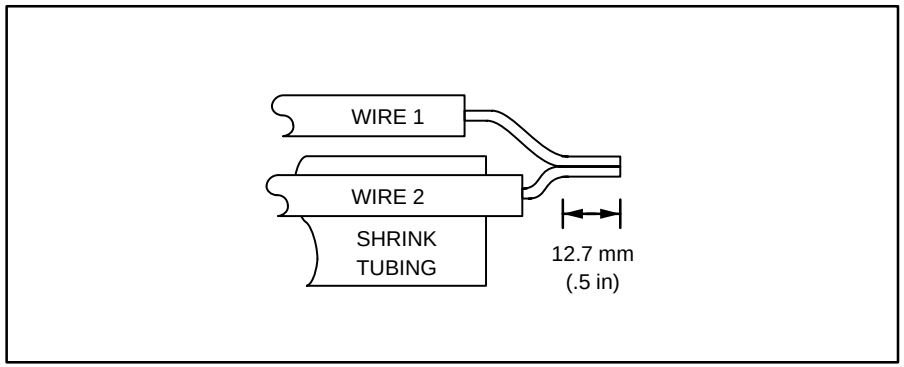
1 = Wiring harness tape
2 = Wiring strand
3 = Broken strands intermittent signal
4 = Circuit insulation

Remove the tape and flex/feel each circuit for a reduction in diameter at break.

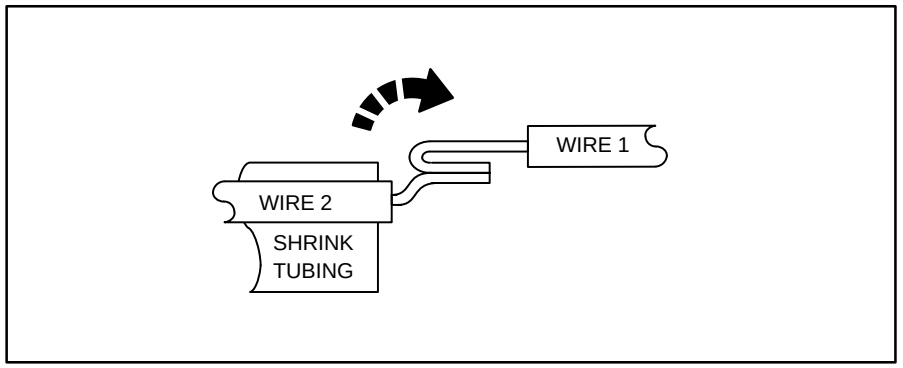
**Recommended splicing method**



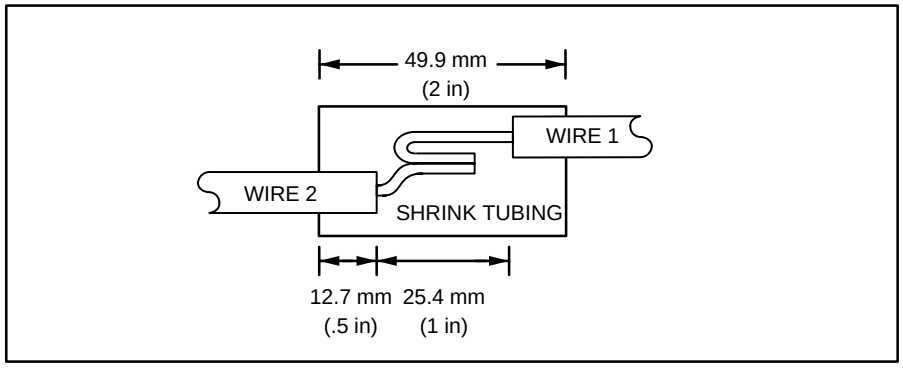

1. Disconnect battery ground cable.
2. Strip wires to appropriate length.



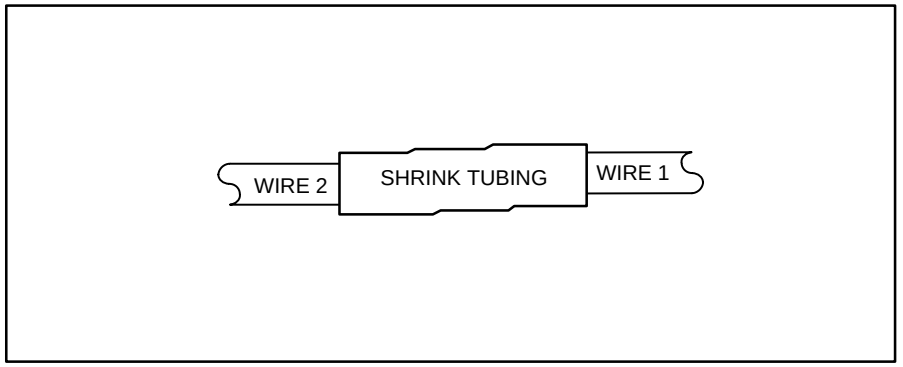

3. Install heat shrink tubing.
4. Twist wires together.
5. Solder wires together.
   **NOTE:** Use rosin core mildly-activated (RMA) solder. Do not use acid core solder.




6. Bend Wire 1 back in a straight line.
   **NOTE:** Wait for solder to cool before moving wires.




7. Evenly position heat shrink tubing over wire repair.
   **NOTE:** Overlap tubing on both wires.




8. Use heat gun to heat the repaired area until adhesive flows out of both ends of heat shrink tubing.
9. Reconnect battery ground cable.